

YUVARAJ S

B.Tech - Artificial Intelligence and Data Science

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CAREER OBJECTIVE

B.Tech student in Artificial Intelligence and Data Science with hands-on experience in Generative AI, LLM, RAG, and Agentic AI development using Python. Skilled in multi-agent workflows, semantic search, multimodal retrieval, and Text-to-SQL systems, with proficiency in LangChain, LangGraph, CrewAI, FastAPI, ChromaDB, and MongoDB. Strong grasp of prompt engineering, NLP, vector embeddings, and API integration. Seeking roles as an AI Engineer, Generative AI Engineer, or Machine Learning Engineer.

TECHNICAL SKILLS:

Programming Languages: Python, Java, C

ML/DL: Machine Learning, Deep Learning, Neural Networks

Generative AI: LLM, Prompt Engineering, RAG Pipelines, Multimodal Retrieval, NLP, Agentic AI & Multi-Agent Systems

Frameworks & Tools: LangChain, LangGraph, CrewAI, FastAPI, Git, GitHub

Databases: MySQL, MongoDB

Vector Search: ChromaDB, Vector Embeddings, Semantic Search, Multimodal Retrieval

Web Technologies: HTML, CSS, JavaScript (Basic)

Core CS: Data Structures and Algorithms, OOPS, DBMS

PROJECTS:

PlacementPal – AI-Powered Placement Preparation Platform

- Built a multi-agent AI platform using LangGraph, CrewAI, and LangChain to generate personalized study plans and company-specific interview insights.
- Implemented a RAG pipeline with ChromaDB for semantic document retrieval, backed by FastAPI and asynchronous MongoDB persistence.
- Integrated the Tavily Search API for real-time company and hiring insights within a two-phase, stateful LangGraph workflow.

Technologies: Python, LangGraph, CrewAI, LangChain, ChromaDB, FastAPI, MongoDB, Tavily Search API, RAG

CropCopilot – AI-Powered Agricultural Intelligence System

- Integrated NVIDIA NIM LLM APIs with CrewAI agents to combine RAG-based knowledge retrieval with Text-to-SQL querying.
- Built and deployed a FastAPI backend serving real-time AI-generated insights with robust API and model error handling.
- Developed a proof of concept combining structured database queries with unstructured knowledge retrieval for agricultural decision support.

Technologies: Python, LLM APIs, CrewAI, RAG, Text-to-SQL, FastAPI

Visual Search Engine for Jewelry using Multimodal RAG

- Built a Retrieval-Augmented Generation pipeline integrating LLM APIs to generate product descriptions from retrieved jewelry data
- Designed vector similarity search across image, sketch, and text inputs for accurate multimodal retrieval
- Structured prompts and parsed LLM outputs to produce consistent, structured product descriptions

Technologies: Python, LLM APIs, RAG, Prompt Engineering, Vector Embeddings, Vector Similarity Search, Multimodal Retrieval

Voice-Based AI Study Assistant

- Developed a voice-enabled AI study assistant capable of analyzing spoken input and extracting study topics automatically.
- Implemented interactive quiz and assessment features to improve learning engagement.
- Enabled real-time audio communication using speech processing techniques.

Technologies: Python, Speech Processing, Speech-to-Text, Text-to-Speech, LLM APIs, Conversational AI

EDUCATION

Bachelor of Technology (B.Tech) – Artificial Intelligence and Data Science

Bannari Amman Institute of Technology, Sathyamangalam, Erode

2023 – 2027 | CGPA: 6.52/10

Higher Secondary Education

Government Higher Secondary School, Vathalmalai, Dharmapuri

2022 – 2023 | 70%

CERTIFICATIONS

Prompt Engineering — Navigate Labs

- Completed a 4-week intensive hands-on training covering prompt design techniques, few-shot and chain-of-thought prompting, and structuring inputs for reliable LLM outputs.

Retrieval-Augmented Generation (RAG) — Navigate Labs

- Completed a 4-week intensive hands-on training covering RAG pipeline design, vector embeddings, and semantic retrieval for grounding LLM responses in external knowledge sources.

Voice Artificial Intelligence — Navigate Labs

- Completed a 4-week intensive hands-on training covering speech-to-text and text-to-speech pipelines, real-time audio processing, and building voice-driven conversational AI systems